

Exploring the relationship between empowerment in using artificial intelligence for problem-solving and artificial intelligence ethical awareness: Multi-group structural equation modelling

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ABSTRACT

While increasing attention has been given to cultivating students' artificial intelligence (AI) ethical awareness, the factors that contribute to its development remain underexplored. Psychological empowerment is a crucial motivational factor influencing students' intentions to incorporate ethical norms in developing AI-based solutions. Hence, this study filled this up by exploring the relationship between empowerment in using AI for problem-solving and AI ethical awareness. Empowerment in using AI for problem-solving in this study comprised three components: impact, self-efficacy and meaningfulness. Data was collected from 681 students from secondary schools and a university in Hong Kong. Structural equation modelling (SEM) results revealed that the impact of using AI for problem-solving positively predicted human autonomy, beneficence, and fairness components of ethical awareness. Meaningfulness in using AI for problem-solving was positively associated with beneficence. However, self-efficacy in using AI for problem-solving negatively predicted beneficence. Multi-group SEM results revealed that gender significantly moderated the structural paths between impact/self-efficacy/meaningfulness in using AI for problem-solving and human autonomy. Such promising findings highlight that psychological empowerment is an effective intervention to cultivate students' ethical awareness in using AI for problem-solving, with the project-based learning approach providing a supportive environment for AI applications.

1. Introduction

With the advancement in AI in various domains such as language learning, healthcare training, visual and voice recognition, and decision-making, educational institutions and scholars worldwide have recently explored the applications of AI in education through multiple forms (e. g., computer programs, applications, embedded control systems in equipment, or robots) to deliver personalised guidance for students and promote their motivations, engagement, and performance (Chu et al., 2025; Hwang et al., 2020; Lai & Tu, 2024; Tu & Lu, 2025). Over time, critics have warned that the expanding use of AI technologies raised a range of ethical concerns, including digital hegemony in education, power relationships among learners, teachers, and AI systems, digital divide, and bias and discrimination against marginalized groups (Allison et al., 2025; Hwang et al., 2020; John-Mathews et al., 2022; Kaur et al.,

2022). AI ethics has consequently attracted increasing attention from scholars in both computer science and education as a crucial component of cultivating citizens' AI literacy (Jang et al., 2022; Ng et al., 2024). It underscores students' understandings and attitudes toward ethical principles (e.g., human autonomy, transparency, fairness, beneficence, accountability, etc) underlying AI technology, thereby facilitating the responsible use, development, and creation of AI technology (Kong, Cheung, & Tsang, 2024).

Current research on AI in education primarily focuses on learners' perceptions of AI use, the development of AI-based learning systems, evaluations of the effectiveness and roles of AI in learning, AI-based learning approaches and strategies, and human-AI interactions (Hwang et al., 2020; Tu & Lu, 2025). Studies on developing ethical principles and practices for employing AI technologies in secondary and higher education (Usher & Barak, 2024; Zhang et al., 2023) and

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designing relevant instruments also increased (Jang et al., 2022; Wang et al., 2025). However, little research has examined what factors can promote the development of students' AI ethical awareness.

Past studies showed that motivational factors (e.g., psychological empowerment) play a crucial role in promoting AI ethics education (Bairaktarova & Woodcock, 2017; Ebadi & Amini, 2024). As Chu et al. (2025)'s study demonstrated that, students in the high learning motivation (HLM) group exhibited significantly greater critical thinking awareness when using AI for creative work than those in the low learning motivation (LLM) group, particularly with regard to ethical concerns, extensive modification of GenAI-generated images, issues related to data sources and accuracy, difficulties in expressing emotion and uniqueness, and the generation of outputs that do not meet expectations.

From the conceptual perspective, empowerment in using AI for problem-solving is motivational-based concept that synthesises meta-cognitive and affective dimensions of AI literacy (Kong et al., 2025). This notion emphasises that empowered students tend to perceive the impacts of using AI for problem-solving in the social context, feel competent to harness the power of AI for problem-solving, and find using AI for problem-solving meaningful. According to the well-known theories underpinning the application of AI technology and ethical awareness, impact and meaningfulness in using AI for problem-solving are parallel to attitudes, subjective task values, and performance expectancy, while self-efficacy in using AI for problem-solving is more associated with perceived behavioural control, expectancies for success, and effort expectancy (Ajzen, 1991; Eccles & Wigfield, 2020; Venkatesh et al., 2003).

These three components are vital mechanisms in facilitating individuals to undergo ethical intentions (e.g., placing ethical concerns (human agency and oversight, beneficence, and fairness) ahead of other concerns (e.g., efficacy, AI-led decision-making, discrimination) before performing ethical behaviours (e.g., AI-based solutions) in practical situations (Ross, 2002). For instance, users who perceive the use of AI for problem-solving as cost-effective and impactful are more likely to benefit from AI support and are less likely to experience ethical anxiety or psychological discomfort (Dholakia, 2001; Gansser & Reich, 2021). Individuals with high self-efficacy tend to persevere when facing ethical dilemmas and adversity, whereas those with lower self-efficacy are more prone to engage in ethical misconduct or decision-making that may negatively impact organisations (Mumford et al., 1993).

Creating the facilitation conditions (e.g., the PBL approach) and strong subjective norms (e.g., perceived social pressure) where students have access to AI-based tools, immediate feedback, and are guided by professional instructors foster individuals' intention to consider ethical issues associated with AI application (Ajzen, 1991; Venkatesh et al., 2003). In doing so, this study adopts a project-based learning (PBL) design within an AI literacy course through a series of student-centred and collaborative activities (e.g., brainstorm a problem, work as a team, uphold inquiry, make decisions together, etc.) (Hamid et al., 2023; Hu, 2024; Saengrith et al., 2022).

The PBL approach supports students in experiencing the solution of real-life problems at the cognitive, affective, and social levels, through which they can observe how important others (e.g., peers, seniors, co-workers) perform and consider ethical principles with AI tools through group work. Furthermore, the literature documented that both practical experience (e.g., problem-solving) and mentorship play critical roles in improving students' understanding of ethical concepts and principles, thereby enabling them to make sound ethical judgments and develop their ethical awareness until morality is formed (Begley, 2006; Corr  a et al., 2024).

Upon these considerations, this study first explored the relationship between empowerment in using AI for problem-solving and AI ethical awareness in Hong Kong students. In addition, gender bias remains a significant concern in AI-related domains due to factors such as algorithmic design, programmer bias, and biased training datasets (Cirillo et al., 2020; Hall & Ellis, 2023; Jiang et al., 2024), and there are gender

disparities in terms of students' understandings of psychological empowerment and prioritisation of AI ethical principles (Eylon & Bamberger, 2000; Singh, 2021). Hence, we also examined how gender moderated the associations between empowerment in using AI for problem-solving and AI ethical awareness in Hong Kong secondary and university students.

2. Literature review

This section begins with an introduction to the definitions of empowerment in using AI for problem-solving and AI ethical awareness. We then proposed that empowerment in using AI for problem-solving and AI ethical awareness are theoretically connected based on three classic models: The theory of planned behaviour (TPB), the situated expectancy-value theory (SEVT) and the unified theory of acceptance and use of technology (UTAUT), all of which have been widely employed as theoretical lens in technology adoption and diffusion research to examine users' intention and behaviour (e.g., Bairaktarova & Woodcock, 2017; Jain et al., 2022; Ma et al., 2025; Xu et al., 2025). Afterwards, we provided some recent research about the associations between empowerment in using AI for problem-solving and AI ethical awareness. Given that gender bias persists in AI algorithms, psychological empowerment, and their ethical understanding, we also explored the moderating role of gender in the relationships of the primary variables.

2.1. Empowerment in using AI for problem-solving

Empowerment in using AI for problem-solving synthesises the meta-cognitive and affective dimension under the AI literacy framework, underlining educated citizens' psychological readiness to leverage AI technology in using AI for problem-solving at the affective levels (Kong et al., 2025; Kong et al., 2024). Empowerment has been a popular concept in management and psychological literature, underlining the experience of being empowered serves as a mediator between empowering managerial practices and outcomes anticipated from empowered employees, such as job performance and innovation (Menon, 1999; Spreitzer, 1995; Thomas & Velthouse, 1990). Besides, empowerment is always considered a multi-dimensional construct that reflects different facets of being psychologically enabled.

For example, Thomas and Velthouse (1990) conceptualised empowerment as increased intrinsic motivation in the workplace with four cognitive components: sense of impact, competence, meaningfulness, and choice. Spreitzer (1995) extended this framework to describe psychological empowerment as a motivational construct with four cognitions that facilitate behavioural outcomes: meaning, competence, self-determination, and impact. Menon (1999) also provided a working definition of psychological empowerment: a cognitive state characterised by a sense of perceived control, competence, and goal internalisation. Subsequently, Frymier et al. (1996) proposed that learner empowerment is composed of impact, competence, and meaningfulness in the instructional context, where the dimension of self-determination was excluded because students, unlike workers, have less willingness and power to expect or exercise self-determined activities in the classroom that are rather predominated by the instructors.

Building on the existing literature on empowerment and AI, Kong et al. (2025) proposed that empowerment in using AI for problem-solving comprised three components: impact in using AI for problem-solving, self-efficacy in using AI for problem-solving, and meaningfulness in using AI for problem-solving. Impact in using AI for problem-solving refers to students' perception of the extent to which their behaviour (e.g., using AI) shapes the task outcomes (e.g., solving problems in real-life contexts) to generate intended strategic, operational, or administrative effects within the situated environments. Self-efficacy in using AI for problem-solving refers to one's perceived competence in using AI technology for solving problems.

Meaningfulness in using AI for problem-solving is students' perceived value of using AI for problem-solving based on their beliefs, ideas, and standards. Thus, empowered students tend to perceive the impacts of using AI for problem-solving in the social context, feel competent to perform it, and find using AI for problem-solving meaningful.

2.2. AI ethics

There is a growing discussion of AI ethics in educational settings, as advancements in AI also bring a diversity of ethical issues in human life, such as social biases or discrimination, invasion of privacy, security attacks, and copyright infringement (Kong & Zhu, 2025; Usher & Barak, 2024). Besides, AI is considered an interdisciplinary field with more social and biological attributes than general digital technology, so students should have social logic rather than machine logic to use, evaluate, and interact with AI (Wang et al., 2023). AI ethics constitutes the ethical dimension of AI literacy, which is defined as human-centred considerations (e.g., fairness, accountability, transparency, ethics, safety) when designing and using AI technology (Ng et al., 2024). Jang et al. (2022) conceptualise AI ethics as students' attitudes to favour or disfavour towards ethical principles like transparency (explainability), fairness, privacy, responsibility, and non-maleficence.

Outside the framework of Ross's (2002) decision-making theory, people have intuitive moral judgment abilities and can decide how to act in practical situations. This suggests that whether to use generative AI products is influenced by the development of their ethical awareness (Schwartz, 2016). Thus, Kong et al. (2024) proposed that AI ethics comprises the social dimension of the AI literacy framework and connected it with problem-solving. In this sense, AI ethics is defined as how educated citizens consider the ethical principles (e.g., human autonomy, benevolence, and fairness) related to AI-based solutions. This perspective emphasises students' intentions or dispositions to consider ethical principles through practical engagement with AI technologies. This notion is also in line with the decision-making theory that individuals experience ethical intentions—such as prioritising ethical concerns (e.g., human agency and oversight, beneficence, and fairness) over competing considerations (e.g., efficacy, AI-led decision-making, discrimination) prior to demonstrating ethical behaviour (e.g., AI-based solutions) in real-world contexts (Ross, 2002).

2.3. Theoretical foundation

2.3.1. The theory of planned behaviour (TPB)

According to the theory of planned behaviour (TPB), perceived behaviour control, and attitudes are two crucial independent antecedents of behaviour intentions (e.g., AI ethics for problem-solving), influencing how hard an individual is willing to attempt to perform the behaviour and how much effort individuals are willing to dedicate towards performing it (Ajzen, 1991; Bairaktarova & Woodcock, 2017). Perceived behavioural control is defined as the amount of control an individual feels they have over a particular situation or behaviour, which is framed as subjective assessment of one's competence in using AI for problem-solving. The second precursor is a social factor termed subjective norm, referring to the perceived social pressure they perform or not to perform a behaviour. Attitudes are the degree to which an individual has a positive or negative evaluation of a particular behaviour, which is termed as students' perceived impact and meaningfulness in using AI for problem-solving. Thus, it is expected that an individual's attitude toward the behaviour (impact/meaningfulness in using AI for problem-solving) and their perceived behavioural control (self-efficacy in using AI for problem-solving) over performing the behaviour strongly predict their behavioural intentions to act in an ethical manner.

2.3.2. Situated expectancy-value theory (SEVT)

As a multi-dimensional motivational construct, empowerment in using AI for problem-solving is built on the synthesis of metacognitive

and affective dimensions of AI literacy (Frymier et al., 1996; Kong et al., 2024). Situated expectancy-value theory can be a prominent theoretical framework to explore how motivational factors and supportive environments facilitate AI ethics education (Eccles & Wigfield, 2020). The SEVT consists of two motivational components, termed students' expectancies for success and subjective task values, which show strong predictive power in explaining individuals' learning intentions, choices of task and activity, and effort and engagement in the chosen activities (Wigfield & Eccles, 2000). Furthermore, prior research has also demonstrated the importance of motivational factors in shaping students' intentions to use AI technology and make ethical considerations (Ebadi & Amini, 2024; Kong, Yang, & Hou, 2024; Zheng et al., 2024).

In the context of using AI for problem-solving, expectancies for success are operationalised as the beliefs about one's competence (e.g., perceived competence) to succeed in using AI for problem-solving, namely, self-efficacy in using AI for problem-solving (Bandura, 1989; Frymier et al., 1996; Wigfield & Eccles, 2000). Subjective task values are referred to as the beliefs about whether a task (e.g., using AI for problem-solving) is impactful, worthwhile, meaningful to engage based on their beliefs, ideas, and standards, yielding intended strategic, administrative, or operating effects of task environments (Frymier et al., 1996; Thomas & Velthouse, 1990; Wigfield & Eccles, 2000). We inferred that one's intention to consider ethical concerns would rely on individuals' expectancies of using AI for problem-solving or their beliefs regarding the values of using AI for problem-solving (e.g., intrinsic value, attainment value, utility value, and cost) and as well as the perceived meaningfulness of such use in terms of inherent rightness or wrongness and perceived fairness or unfairness (Jung, 2009).

2.3.3. The unified theory of acceptance and use of technology (UTAUT)

The unified theory of acceptance and use of technology (UTAUT) model developed by Venkatesh et al. (2003) suggests that four core constructs (performance expectancy, effort expectancy, social influence and facilitating conditions) are direct determinants of behavioural intention and that these factors are in turn moderated by variables like gender, age, experience, and voluntariness of use. A large amount of research also supported that the effectiveness of the UTAUT model in explaining and predicting users' intentions and behaviours with Generative AI Products, such as ChatGPT, Chatbot, and AI-enabled language e-learning systems (Jain et al., 2022; Ma et al., 2025; Xu et al., 2025). Recent studies have also extended the UTAUT model to explore students' intentions of generative AI products from an ethical perspective (e.g., Kajiwaru & Kawabata, 2024; Zhu et al., 2025).

Performance expectancy refers to the extent to which individuals believe that the use of new technologies can help them achieve better performance in the study (Venkatesh et al., 2003). It may be analogous to the impact and meaningfulness of using AI for problem-solving under the empowerment construct. In this vein, students who tend to perceive the benefits and positive impacts through using AI technology to solve issues, such as personalised learning and personalised therapeutic interventions, career preparation, psychological counselling, and automated writing evaluation, are more likely to consider the ethical principles when using AI technology for solving problems. Effort expectancy is described as the perceived difficulty of using new technologies, emphasising students' perceived effort and time to invest in using AI technology for problem-solving (Strzelecki, 2024; Venkatesh et al., 2003). In this study, it is parallel to self-efficacy in using AI for problem-solving. Therefore, we anticipated that the easier students find it to use AI for problem-solving, the more willing they will be to consider its potential ethical influence.

2.4. Relationship between empowerment in using AI for problem-solving and AI ethical awareness

Digital technologies mediate most human experience (Calvo et al., 2020, pp. 31–54). The relationship between AI technology and ethical

behaviour cannot be accomplished simply by using regulations, ethical guidelines, or principles (Johnson, 2004). Specific training seminars or a standalone semester course, where students could obtain the practice of appropriate behaviours in line with the professional code of ethics, is also necessary to help students resolve ethical challenges and improve their ethical sensitivity (Mahdiyoun et al., 2017). Bagherzadeh et al.'s (2021) quasi-experimental design research demonstrated that the empowerment course serves as a vital mechanism for developing students' ethical awareness by providing lectures on the principles of professional ethics, the importance and application of ethical concepts, as well as training students' capacities in ethical decision-making and resolving ethical problems and conflicts.

Another study by Abbasi et al. (2019) demonstrated that an empowerment program, through providing examples of ethical skills, exchanging ethical experience, and strengthening problem-solving skills, alleviates individuals' ethical distress and improves their performance in their field. Kong et al. (2023) designed a course that involves the cultivation of secondary students' empowerment in AI-related tasks by using AI to resolve real-life problems, finding that they felt more empowered in AI-based solutions with enhanced ethical awareness. Similarly, Kajiwara and Kawabata (2024) indicated that secondary school students are more likely to understand how to use AI technology ethically when exposed to a Chatbot for decision-making. Specifically, approximately 70% of students prioritise the ethical use of data and adhere to the principle of non-maleficence, while only 8.7% reported that they would rely on ChatGPT's responses as their definitive decisions.

Users are more likely to adopt technology when it is both effective and cost-efficient for them to use (Gansser & Reich, 2021). Such positive recognition of the impact of AI-based technology can increase students' intention to embed ethical values and norms into computer programs to support privacy or property rights instead of against realising these values (Floridi & Cows, 2019). Students reported favourable anticipations for using AI for creative support, decision support, and information retrieval (Kajiwara & Kawabata, 2024). However, the perceived risks associated with these anticipated behaviours may lead to ethical anxiety or psychological discomfort (Dholakia, 2001). Ashok et al. (2022) indicated that though AI technology has yielded profound impacts on technological and societal development, the use of AI involves resolving ethical implications in the physical [e.g., safety and sustainability], cognitive [e.g., autonomy, fairness, and prosperity], information [e.g., privacy], and governance [e.g., financial, social, and individual impact] domain.

The growing use of AI technology also intensified discussions regarding the negative impacts of unfairness and bias on individuals and society, such as amplifying existing inequalities, perpetuating stereotypes and discrimination against marginalized groups, limiting individuals' access to essential services, entrenching discriminatory narratives and hindering efforts toward equality and inclusivity (Ferrara, 2023; Hall & Ellis, 2023; Jiang et al., 2024). This is particularly obvious in areas like healthcare, employment, criminal justice, credit scoring, and synthetic media.

Self-efficacy becomes particularly useful to predict ethical behaviors in an AI-saturated world since the ethical issues behind AI technology remain novel and ambiguous to the vast majority of the public, who lack AI education experience or technical backgrounds to effectively engage with AI, collaborate with it, and critically assess its applications (Jang et al., 2022; Long & Magerko, 2020). According to social cognitive theory, self-efficacy is also a crucial factor in self-regulation, which has formidable predictive powers for more effective management of human ethical performance (Bandura, 1989). Research indicated that self-efficacy influences students' cognitive processes and perceptions involved in ethical decision-making, which, in turn, could shape ethical behaviours and attitudes toward ethical misconduct (Elias, 2009; MacNab & Worthley, 2008; Stenmark et al., 2021). Besides, people with high self-efficacy are more likely to accept difficult tasks and persevere when

encountering adversity, whereas those with lower self-efficacy tend to resort to ethical misconduct or make decisions that are harmful to organisations (Mumford et al., 1993). Using AI also benefits those with higher skills and wages more, since AI applications may disproportionately and adversely impact lower-skilled and lower-paid workers (Ernst et al., 2018).

Therefore, individuals with higher efficacy in using AI for problem-solving (e.g., individuals with formal AI or computer science education experience) are more likely to consider justice concerns around how the benefits and burdens of AI are being distributed with a more positive attitude than those without AI education experience (Banks & Formosa, 2023; Jang et al., 2022; Laupichler et al., 2023). However, those with lower self-efficacy tend to lack sufficient understanding of the algorithms used to train AI models or the rationale behind AI's decisions. This limited understanding may diminish their sense of responsibility to ensure the beneficence of AI-based solutions on society and disrupt the balance between human and AI decision-making (Floridi & Cows, 2019).

More recently, some researchers have also linked ethical concerns surrounding AI with the meaningfulness of work, particularly concerning sustainability and AI's impact on fellow humans and future generations (Nyholm & R  ther, 2023; R  ther, 2024). This is because providing opportunities for meaningful work supports positive outcomes for AI users and is ethically crucial as a foundation for human well-being and flourishing (Bailey et al., 2019; Lysova et al., 2019). Thanks to AI systems, people can perceive transformative advancements across vital sectors brought by AI to make their daily lives more manageable and sustainable (Taddeo & Floridi, 2018). These developments may enable humans to perceive the meaningfulness of AI applications in relation to their characteristics, interests, potential abilities or skills, aspirations, and life projects.

2.5. Connecting empowerment in using AI for problem-solving and AI ethical awareness: the project-based learning (PBL) approach

The recent SEVT theory also emphasised how supportive environments play a critical role in influencing students' behavioural intentions (Eccles & Wigfield, 2020). Another crucial factor is subjective norm, which refers to individuals' beliefs about whether people they care about would support them performing a behaviour (Ajzen, 1991). Similarly, two contextual attributes of the UTAUT model, named as facilitating conditions and social influence, underscore that access to technical resources and supportive environment, as well as social influence (e.g., peers, seniors, co-workers, and supervisors), also influence individual intentions to consider using AI technology ethically (Venkatesh et al., 2003).

Zhu et al. (2025) indicated the importance of facilitating conditions for AI education. It is challenging for university students in the early stages to consider ethical issues if there are objective constraints hinders (e.g., software and hardware support, user guides, tutorials, and supporting facilities) them to use AI products in practice. Besides, previous studies found that supportive environments or facilitating conditions and subjective norm positively affect performance expectancy and effort expectancy (Dwivedi et al., 2019; Jain et al., 2022; Shaw & Sergueeva, 2019; Tseng et al., 2022). In the organisational structure, facilitating conditions like providing training opportunities, positive attitudes toward ethical standards, and reasonable expectations can significantly enhance individuals' adherence to ethical codes (Bordbar et al., 2025; Dehghani et al., 2017).

On the other hand, secondary and university students typically receive guidance from teachers, schools and universities and hope to obtain their recognition and appreciation, and AI-enabled tools are always used by teams to manage work (Nasaj et al., 2025; Sohn & Kwon, 2020). Therefore, significant others (e.g., teachers and peers) have a prominent effect for students to use generative AI products more actively in compliance with ethical norms (Tseng et al., 2022; Wang &

Hazari, 2018). Also, ethical development, grounded in Aristotle's conception of virtues, is promoted by both practice and the role of a mentor to facilitate students' understanding of ethical frameworks, thereby aiding them in making sound ethical judgments in practice (Begley, 2006).

The PBL approach has been widely applied in AI training courses by providing supportive environments in which learners actively seek knowledge to solve defined problems through group collaboration and available AI-based tools, and the teacher serves as a supporter and facilitator to enhance their analytical thinking and problem-solving skills (Almazroui, 2023; Hamid et al., 2023; Hu, 2024). In the PBL context, students are presented with real-world case studies or scenarios that simulate the challenges in their daily life that could be solved with AI technology. This engages students in a collaborative and student-centred learning process where students define problems reflecting real-life experiences, work as a team, solve problems collectively, and make decisions together before presenting their projects (Amiruddin et al., 2025; Williams et al., 2023). It is an effective approach that empowers students to apply their knowledge, skills, and attitudes meaningfully and practically and helps them integrate the work responsibilities and problem-solving skills within a constantly changing environment or a new scenario (Kim et al., 2016).

Saengrith et al. (2022) embedded problem-based blended training via chatbot to enhance problem-solving skills in the workplace, finding that students' problem-skills were higher significantly improved after the training, with high levels of skill retention remained through the subsequent implementation phases: 1) Group identification; (2) Problem identification; (3) Idea creation; (4) Learning; (5) Implementation; (6) Evaluation. Hamid et al. (2023) also integrated ChatGPT into a project-based learning session to ensure that ChatGPT's benefits do not undermine creative and critical thinking capabilities. The results indicated that the PBL encourages a thorough understanding of the subject matter, challenges the accuracy of the presented information brought by ChatGPT, and replicates real-world scenarios; thereby strengthening problem-solving and critical thinking skills.

Furthermore, numerous studies have regarded the importance of the PBL approach in fostering students' AI ethical awareness through engaging them in addressing real-life ethical issues, thereby fostering a meaningful blend of theoretical understanding and practical experience of ethical principles (Corrêa et al., 2024; Hu, 2024). Almazroui (2023) indicated that the PBL is powerful in ethical education in the UAE since it allows students to construct knowledge in solving problems, design and implement projects that reinforce their appreciation of ethical values, and cultivate specific values aligned with the UAE's predominantly group-oriented culture. Similarly, Williams et al. (2023) embedded ethics education into three project-based curricula, including *Creative AI*, *Dancing with AI*, and *How to Train Your Robot*. The findings showed that hands-on unplugged and programming activities facilitated students' understanding of AI and enabled students to develop a critical lens to understand better how AI systems work and impact society.

Hence, this study employed the PBL approach to develop students' empowerment in using AI technology to resolve real-life problems, accompanied by cultivating their AI ethical awareness. Through PBL, it first provides supportive environments for students to leverage AI power and address real ethical concerns when developing AI-based solutions, contributing to narrowing the principle-practice gap of AI ethics because it is a developmental and ongoing process (Bordbar et al., 2025; Corrêa et al., 2024). In addition, students could perceive how their significant others (e.g., peers or teachers) perceive the ethical use of AI technology, motivating them to develop a stronger intention to consider ethical concerns (Begley, 2006; Nasaj et al., 2025; Venkatesh et al., 2003).

2.6. Gender differences in empowerment in using AI for problem-solving and AI ethical awareness

A growing body of literature has shown that gender bias in AI

algorithms is a vital concern, which has recently prompted the fairness discussion of machine learning and AI recently (Hall & Ellis, 2023; Jiang et al., 2024; Shrestha & Das, 2022). Gender bias can be derived from algorithmic design and training datasets, programmer bias, and gender stereotypes (Cirillo et al., 2020; Hall & Ellis, 2023; Nadeem, Abedin, & Marjanovic, 2020). AI technology is often constructed as a male domain composed of male attributes such as logic, structure and technical knowledge, while women are frequently underrepresented in AI training datasets and AI development teams (Hall & Ellis, 2023; Sultan et al., 2019). Thus, few women have the confidence and support to pursue computer-related fields, particularly in AI, with a mere 16% of AI-focused tenure-track faculty being women in 2021 and 26% of Data Science and AI occupations held by women (Diekmann et al., 2017; Jiang et al., 2024; Zhang et al., 2021).

Programmer bias, whether conscious or unconscious, seeps into AI algorithms. For example, Amazon's and LinkedIn's AI-based resume screening systems were found to preferentially send information technology (IT) job postings to male candidates, and some AI models automatically associate male pronouns (he/him) to professions such as physicians and pilots while assigning female pronouns (she/her) to traditionally female-dominated roles like nurses and flight attendants (Shrestha & Das, 2022; West et al., 2019). In addition, there are long traditions of gender stereotypes related to masculine instrumentalism and feminine expressiveness, which have been perpetuated in the algorithm and eventually amplify the bias in society (Faulkner, 2000; Nadeem et al., 2020). However, some research demonstrated that marginalized female students also demonstrated creativity in developing applications for addressing real-world problems and community needs, who can leverage their unique strength to learn computing concepts to promote their success and advancement in the field of AI through their diverse lived experiences and innovative solutions (Pinkard et al., 2017; Searle & Kafai, 2015; Vrieler & Salminen-Karlsson, 2022).

On the other hand, researchers have argued that prevailing conceptualisations of empowerment may reflect a masculine bias as they emphasise attributes such as competence, mastery, and control, while overlooking more feminine-oriented dimensions such as communion, connectedness, and cooperation (Alimo-Metcalfe, 1995; Eylon & Bamberger, 2000; Riger, 2002). Moreover, men are more likely to define or express power as control over their environment, which may make "impact" become the most salient component of empowerment in predominantly male samples. Inversely, Abukhait et al. (2019) demonstrated that women exhibited higher levels of empowerment than men. This outcome may reflect their greater willingness to engage in knowledge-sharing behaviours, enhancing their innovation capacity in male-dominated societies.

Gender identities also influence the prioritisation of AI ethical principles (Roxas & Stoneback, 2004; Singh, 2021). Male users tend to emphasise autonomy, accountability, and transparency in AI systems, reflecting their preference for independent control during human-AI interactions and understandable and responsible technological practices (Singh, 2021). In contrast, females do not emphasise these aspects but prioritise privacy and data security due to their heightened concerns about data security and implications, as well as protective measures related to technology usage (Singh, 2021). Given that gender gaps persist in the AI-related workplace and research, females are more concerned about fairness and non-discrimination to increase their inclusion and performance in AI activities (Jiang et al., 2024; Singh, 2021).

In addition, women reported higher levels of perceived severity and fear of cyber deviance behaviours, which may continue to exist when it comes to AI ethics because of distorted gender-based power relations (Kim & Shin, 2021). Gender differences were also confirmed in students' attitudes toward AI (Jang et al., 2022; Kim & Shin, 2021). Specifically, females demonstrated greater awareness of fairness, privacy, and non-maleficence compared to males, but there were no statistically

significant differences in responsibility and transparency across gender (Jang et al., 2022).

2.7. The present study

The present study investigates the relationship between empowerment and using AI for problem-solving and AI ethical awareness among Hong Kong students (see Fig. 1). Thus, the following research questions and hypotheses were addressed.

RQ 1: To what extent does the impact of using AI for problem-solving predict Hong Kong students' AI ethical awareness? Hypothesis 1: Impact of using AI for problem-solving predicts human autonomy. Hypothesis 2: Impact of using AI for problem-solving predicts beneficence. Hypothesis 3: Impact of using AI for problem-solving predicts fairness.

RQ 2: To what extent does self-efficacy of using AI for problem-solving predict Hong Kong students' AI ethical awareness in Hong Kong students? Hypothesis 4: Self-efficacy in using AI for problem-solving predicts human autonomy. Hypothesis 5: Self-efficacy in using AI for problem-solving predicts beneficence. Hypothesis 6: Self-efficacy in using AI for problem-solving predicts fairness.

RQ 3: To what extent does meaningfulness in using AI for problem-solving predict Hong Kong students' AI ethical awareness? Hypothesis 7: Meaningfulness in using AI for problem-solving predicts human autonomy. Hypothesis 8: Meaningfulness in using AI for problem-solving predicts beneficence. Hypothesis 9: Meaningfulness in using AI for problem-solving predicts fairness.

RQ4: Are there gender differences in the relationship between empowerment in using AI for problem-solving and AI ethical awareness?

3. Methodology

3.1. Research context

This study was based on the third part (Course 3) of the AI literacy programme in Hong Kong. The first two courses developed a conceptual understanding of two crucial AI areas. Only Students who finished the first two courses could proceed to Course 3. The third course consisted of

a set of workshops totalling 14 h that engaged students in developing AI projects to address real-life problems with ethical considerations of their AI-based solutions, with the PBL approach adopted throughout the course. We recruited students from 64 secondary schools and the authors' university in Hong Kong from 2021 onwards to develop their AI literacy (see Table 1). The entire programme was free of charge and non-credit-bearing, which consisted of three courses: Machine Learning, Deep Learning, and Developing Artificial Intelligence Applications (Kong et al., 2023). The instructors of these courses were highly qualified academically, holding either a doctoral degree, a master's degree, or a postgraduate diploma in education.

Before the commencement of Course 3, students were provided with self-learning materials and introductory lectures. These resources illustrated real-life scenarios involving the design and usage of AI, as well as how students consider ethical principles or dilemmas (e.g., human autonomy, beneficence, and fairness) in AI applications. Following the introductory lectures, students formed their groups and prepared for their group projects to solve real-life problems via various AI-powered platforms that were available freely or via institutional subscription.

The specified procedures of the group project were described as follows: (1) Students brainstormed a problem that AI-enabled tools could address, and the instructors then showcased the application of AI-enabled tools, including Google Teachable Machine (for recognizing images, sounds, and postures), Microsoft Azure QnA Maker (for creating chatbots), and Microsoft Azure Machine Learning Studio (for analyzing

Table 1

Distribution of cohorts for secondary school and university students across four years.

	Cohorts	N	Percentage
Secondary Students	2021–2022	151	26.003%
	2022–2023	54	8.202%
	2023–2024	185	30.716%
University Students	2022–2023_sem 1	33	5.314%
	2022–2023_sem 2	45	7.246%
	2023–2024_sem 1	75	12.077%
	2023–2024_sem 2	24	3.864%
	2024–2025_sem 1	55	8.85%
	Total	621	100%

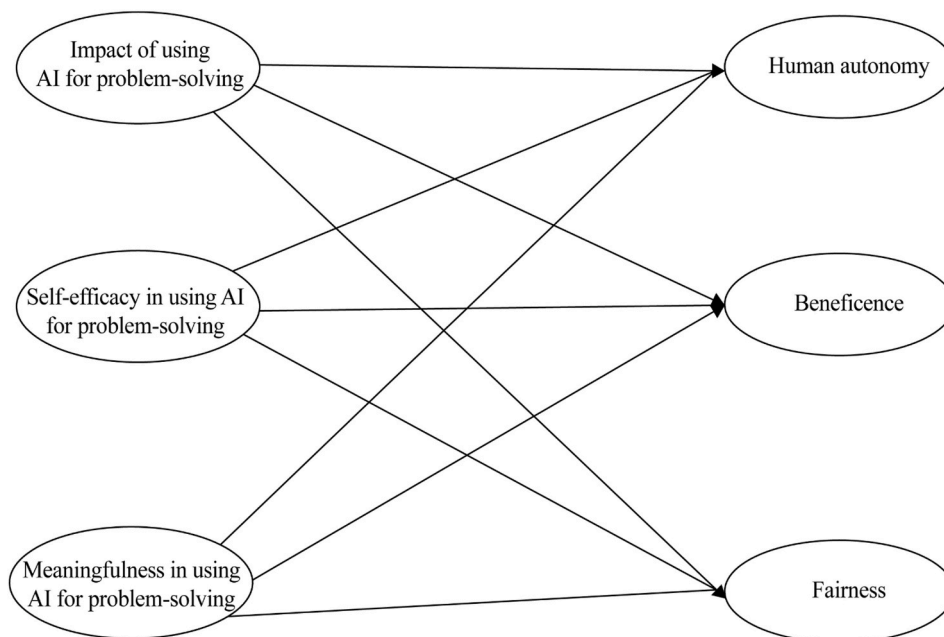


Fig. 1. Conceptual model.

and predicting quantitative data). (2) The instructors organised individual consultations for project development instructions that were built on the five steps of machine learning: (a) problem definition; (b) data collection; (c) data pre-processing; (d) model training; and (e) inference and prediction.

(3) The instructors guided students to consider the ethical implications of their AI-based solutions and explore ways to mitigate any potential ethical concerns that might arise. (4) students participated in 6-h collaborative project sessions to develop their AI-based solutions for their self-defined problems, such as data collection, AI models training for obtaining solutions, reflection on encountered challenges and potential enhancements, and ethical considerations linked to AI-based solutions. (5) Students demonstrated all the stages of the AI problem-solving projects in project presentations, followed by peer evaluations and discussions on the ethical issues involved in each project.

3.2. Participants

A total of 681 students were recruited to participate in the third component (Course 3) of the AI literacy programme through an open call at the authors' university and across secondary schools in Hong Kong. Participants with missing data over 5% and outliers for each of the measures were removed (see Statistical analysis for details), resulting in a final sample size of 621 students remaining for all analyses. From Table 2, 336 (54.106%) were males and 281 were females (45.250%), with 4 students (0.644%) opting not to disclose their gender.

390 (62.802%) students were from 64 secondary schools, of which 319 students were from 47 (73.438%) aided schools, 48 students were from 12 (18.750%) direct subsidy scheme schools, and 20 students were from 5 (7.813%) government schools. 231 university students (37.198%), including 3 who chose not to report their disciplines, who either enrolled in bachelor's degree or higher diploma programs were from diverse disciplines, such as mathematics (8.225%), liberal studies education (7.792%), Chinese language studies (7.359%), artificial intelligence and education technology (5.628%), information and communication technology (5.195%) and so on.

In line with emerging research that highlights the development of AI literacy in audiences without technical backgrounds (Jang et al., 2022; Kong et al., 2023; Laupichler et al., 2023), our programme design also did not require programming knowledge. Accordingly, there were 237 (38.1643%) students reported that they did not have a programme background, in contrast with 371 (59.7424%) students with programming knowledge. The remaining 13 (2.0933%) students did not disclose their programming background.

3.3. Measures

Empowerment in Using AI for Problem-Solving Scale (EUAIPS). The EUAIPS is a 11-item self-report measure including three subscales: *Impact in using AI for problem-solving* (IP, 4 items, #I want to apply my AI

knowledge and skills to solve problems in daily life.), *Self-efficacy in using AI for problem-solving* (SE, 4 items, #I think of myself as someone who can solve problems with AI), and *Meaningfulness in using AI for problem-solving* (MN, 3 items; #Using AI to solve problems is important to me) (Kong et al., 2024). This measure of the EUAIPS uses a 5-point Likert response scale (e.g., 1 = strongly disagree; 5 = strongly agree). In this study, the EUAIPS indicates good reliability with Cronbach's α ranging from 0.907 to 0.960 (see Table 3).

AI Ethical Awareness Scale (AIEAS). A 9-item AIEAS was used to measure students' AI ethical awareness incorporating three subscales of *Human Autonomy* (HA, 3 items, #People involved with AI should be treated as autonomous individuals), *Beneficence* (3 items, #The designers should consider that the benefits brought by AI should outweigh the risks) and *Fairness* (3 items, #AI technologies should promote social justice) (Kong & Zhu, 2025). Its Cronbach's α ranged from 0.802 to 0.921, indicating good internal consistency in this study (see Table 3).

3.4. Data collection

The research ethics were reviewed and approved by a university in Hong Kong (Ref. no. XXXXX-XXXX-XXXX and XXXX-XXXX-XXXX). Participants were recruited from the authors' university and various types of secondary schools in Hong Kong, including government schools, aided schools, and direct subsidy scheme schools. We obtained institutional and student consent prior to data collection. Participants were informed of the research purpose beforehand, and their participation was entirely voluntary, with the option to withdraw from the study at any time.

The entire course spanned four weeks, during which students were required to develop AI applications by completing their self-designed problem-solving projects. At the end of the course, participants were asked to evaluate their empowerment in using AI for problem-solving and their ethical considerations when developing AI projects to solve problems via Qualtrics. They were given 10–15 min to complete the EUAIPS and AIEAS without teachers' supervision during class hours, and report their demographic information (e.g., gender, schools, and disciplines) was also reported.

3.5. Statistical analysis

None of the participants' responses were discarded since there were no item-level missing values (Tabachnick & Fidell, 2007). 60 cases were identified as outliers given that Mahalanobis' d^2 values exceeded the chi-square cutoff scores ($\chi^2(40) = 45.315$, $p = 0.001$), leading to the final sample size of 621 participants for all subsequent analyses. The normal distribution of our data was examined following the guidelines by Kline (2015): the range of skewness (± 3) and Kurtosis (± 10) values was acceptable. Descriptive statistics, internal consistency, and bivariate correlations among the primary variables were performed using SPSS 29. The subsequent analyses were performed within the R statistical computing environment using maximum likelihood estimation (R Core Team, 2020).

We first ran confirmatory factor analysis (CFA) to examine the factor structure of empowerment in using AI for problem-solving and AI ethical awareness. To answer RQ1, RQ2, and RQ3, Structural equation modelling (SEM) was employed to explore how empowerment in using AI for problem-solving related to AI ethical awareness dimensions (see Fig. 1). To answer RQ4, we also investigated how gender moderated the relationship between empowerment in using AI for problem-solving and AI ethical awareness through multi-group SEM analysis. Prior to this, we assessed measurement invariance across gender of the constructs of empowerment in using AI for problem-solving and AI ethical awareness by testing configural, metric, scalar, and strict invariance sequentially (Vandenberg & Lance, 2000).

Table 2
Demographic information for the participants (N = 621).

Parameter	Specification	Frequency	Percentage
Gender	Male	336	54.106%
	Female	281	45.250%
	No reply	4	0.644%
Education level	Secondary school	390	62.802%
	University	231	37.198%
Programming knowledge	Know programming	371	59.7424%
	Do not Know programming	237	38.1643%
	No reply	13	2.0933%
Total		621	100%

Table 3

Descriptive statistics, Cronbach's alpha coefficient, and bivariate correlations between primary variables.

Variables	N	1	2	3	4	5	6
1. Impact of using AI for problem-solving	4	(0.960)					
2. Self-efficacy in using AI for problem-solving	4	0.722***	(0.938)				
3. Meaningfulness of using AI for problem-solving	3	0.827***	0.836***	(0.907)			
4. Human autonomy	3	0.645***	0.541***	0.600***	(0.802)		
5. Beneficence	3	0.683***	0.540***	0.650***	0.742***	(0.832)	
6. Fairness	3	0.734***	0.595***	0.685***	0.707***	0.795***	(0.921)
Mean	—	4.269	4.045	4.181	4.142	4.282	4.236
Standard deviation	—	0.571	0.631	0.582	0.628	0.615	0.631
Skewness	—	- 0.409	- 0.183	- 0.317	- 0.550	- 0.559	- 0.418
Kurtosis	—	0.989	0.263	0.746	1.233	0.387	0.136

Note. Values shown in parentheses on the diagonal are Cronbach's α of the primary variables. *** $p < 0.001$.

4. Results

4.1. Descriptive statistics, internal consistency, and bivariate correlations

Indicated by Table 3, the absolute values of skewness (range: - 0.559 to - 0.183) and kurtosis (range: 0.136 to 1.233) fell within the threshold (Kline, 2015), indicating that the normal distribution of the dataset was met. Cronbach's alpha values for the primary variables ranged from 0.802 to 0.960, suggesting good reliability of each scale. Students reported the highest perception of the “impact of using AI for problem-solving” with the lowest variability ($M = 4.269$, $SD = 0.571$). Meanwhile, they considered the ethical principle of beneficence most frequently ($M = 4.282$, $SD = 0.615$) with a lower level of variability than human autonomy and fairness.

The correlation results showed that the impact of using AI for problem-solving yielded the highest correlation with fairness ($r = 0.734$, $p < 0.001$). Similarly, it was also strongly correlated with human autonomy ($r = 0.645$, $p < 0.001$) and beneficence ($r = 0.683$, $p < 0.001$). Statistically significant and positive correlations were also observed between meaningfulness in using AI for problem-solving and AI ethical awareness dimensions ($r = 0.600$ to 0.685 , $p < 0.001$), which is higher than the correlations between self-efficacy in using AI for problem-solving and AI ethics dimensions, ranging from $r = 0.540$ to 0.595 , $p < 0.001$).

4.2. CFA results

As shown in Table 4, the three-factor model of the EUAIPS fitted the data well: $\chi^2(41) = 165.520$, $p < 0.001$, CFI = 0.985, TLI = 0.979, RMSEA = 0.069 (90% CI [0.058, 0.080]), SRMR = 0.024. The fit indices of the three-factor model of the AILEAS were also satisfactory: $\chi^2(24) = 113.737$, $p < 0.001$, CFI = 0.979, TLI = 0.968, RMSEA = 0.078 (90% CI [0.064, 0.092]), SRMR = 0.025. The CFA results provided strong evidence for construct validity of the EUAIPS and AILEAS, so these two models were used for all subsequent analyses.

Table 4

Confirmatory factor analysis results of the EUAIPS and AIEAS.

Models	χ^2	df	CFI	TLI	RMSEA (90% CI)	SRMR
Empowerment of using AI for problem-solving	162.520***	41	0.985	0.979	0.069 (0.058, 0.080)	0.024
AI ethical awareness	113.737***	24	0.979	0.968	0.078 (0.064, 0.092)	0.025

Note. CFI = Comparative fit index; TLI = Tucker–Lewis index; RMSEA = Root mean squared error of approximation; SRMR = Standardised root mean squared residual; AIC = Akaike information criterion; CFA = Confirmatory factor analysis. *** $p < 0.001$.

4.3. SEM results

The structural model demonstrated a satisfactory fit: $\chi^2(155) = 427.820$, $p < 0.001$, CFI = 0.979, TLI = 0.974, RMSEA = 0.053 (90% CI [0.047, 0.059]), SRMR = 0.028. The SEM model explained about 61.8% variance of beneficence, followed by 61.6% variance of fairness and 53.7% variance of human autonomy, accordingly.

According to Table 5 and Fig. 2, impact of using AI for problem-solving significantly and positively predicted human autonomy ($\beta = 0.461$, $p < 0.001$, [95%CI = 0.206, 0.717]), beneficence ($\beta = 0.417$, $p < 0.01$, [95%CI = 0.164, 0.670]) and fairness ($\beta = 0.509$, $p < 0.001$, [95%CI = 0.284, 0.735]). Thus, H1, H2 and H3 were supported. Unexpectedly, self-efficacy in using AI for problem-solving was a significant negative predictor of beneficence ($\beta = - 0.322$, $p < 0.05$, [95%CI = - 0.608, - 0.035]), with H5 supported. However, we did not find support for the H4 and H6 that self-efficacy in using AI for problem-solving significantly predicted human autonomy and fairness; thereby, H4 and H6 were rejected. We also found that meaningfulness in using AI for problem-solving had a positive prediction on beneficence ($\beta = 0.666$, $p < 0.01$, [95%CI = 0.208, 0.123]), but it did not have a significant effect on human autonomy and fairness. Hence, H8 was supported, whereas H7 and H9 were rejected.

Table 5

Standardised estimates and testing results of the nine hypotheses.

Hypotheses	β	95%CI		Test results
		LL	UL	
H1: Impact of using AI for problem-solving → human autonomy	0.461***	0.206	0.717	Supported
H2: Impact of using AI for problem-solving → beneficence	0.417**	0.164	0.670	Supported
H3: Impact of using AI for problem-solving → fairness	0.509***	0.284	0.735	Supported
H4: Self-efficacy in using AI for problem-solving → human autonomy	0.008	– 0.258	0.273	Rejected
H5: Self-efficacy in using AI for problem-solving → beneficence	- 0.322*	– 0.608	– 0.035	Supported
H6: Self-efficacy in using AI for problem-solving → fairness	- 0.084	– 0.350	0.181	Rejected
H7: Meaningfulness in using AI for problem-solving → human autonomy	0.286	– 0.172	0.744	Rejected
H8: Meaningfulness in using AI for problem-solving → beneficence	0.666**	0.208	1.123	Supported
H9: Meaningfulness in using AI for problem-solving → fairness	0.371	– 0.042	0.785	Rejected

Note. 95% confidence interval (CI) with bootstrapping (bootstrap sample = 5000), standardised estimates. LL = lower limit; UL = Upper limit. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

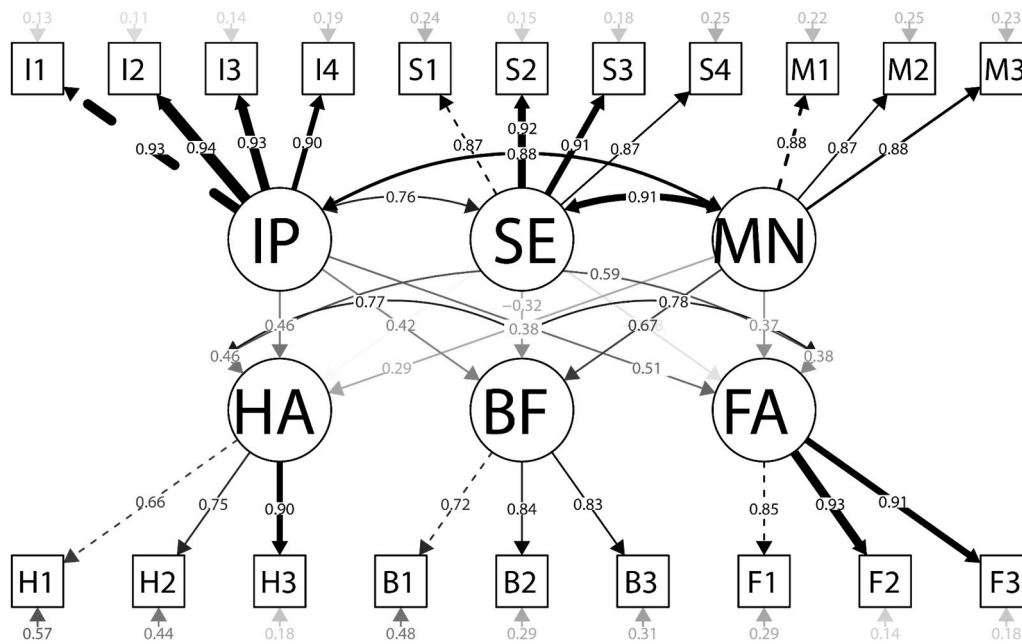


Fig. 2. SEM model of empowerment of using AI for problem-solving and AI ethical awareness. Note. IP = Impact of using AI for problem-solving; SE = self-efficacy in using AI for problem-solving; MN = meaningfulness in using AI for problem-solving; HA = human autonomy; BF = beneficence; FA = fairness.

4.4. Multi-group SEM results

Table 6 summarises the measurement invariance results of the constructs of empowerment in using AI for problem-solving and AI ethical awareness, which is a necessary prerequisite for testing structural invariances in cases where group differences arise from measurement parameters (Vandenberg & Lance, 2000). The baseline model tests configural invariance, which involves the invariance of factor structure across groups without imposing constraints on all parameters, showing a good fit to the data: $\chi^2(310) = 761.702, p < 0.001, CFI = 0.965, TLI = 0.957, RMSEA = 0.069$ (90% CI [0.063, 0.075]), SRMR = 0.035. We then continued to test the metric invariance model by imposing constraints on the factor loadings. The differences between metric and configural invariance were $\Delta CFI = 0.000$ and $\Delta RMSEA = -0.002$, indicating that metric invariance was attained. To evaluate the scalar invariance model, we added additional equality constraints on item intercepts while keeping constraints on the factor loadings. Results indicated that the scalar invariance model was supported, given that the differences in scalar and metric invariance models were $\Delta CFI = 0.000$ and $\Delta RMSEA = -0.001$.

Hence, we tested the multi-group SEM model across gender, which showed a good fit to the data: $\chi^2(338) = 790.328, p < 0.001, CFI = 0.965, TLI = 0.960, RMSEA = 0.066$ (90% CI [0.060, 0.072]), SRMR = 0.039. A graphical representation of the estimated structural model for each group is displayed in Fig. 3. Wald Chi-square tests showed that three structural paths significantly differed across gender (see Table 7). The structural path from the impact of using AI for problem-solving to human autonomy was significant for males ($\beta = 0.629, p < 0.001$) but not for females ($\beta = 0.152, p > 0.05$): $\chi^2(1) = 5.062, p < 0.05$.

An opposite finding could be identified in the path meaningfulness in

using AI for problem-solving to human autonomy; the path coefficient was significant for female students ($\beta = 0.978, p < 0.01$) but not for male students ($\beta = -0.098, p > 0.05$): $\chi^2(1) = 6.243, p < 0.05$. There are also significant gender differences in the relationship between self-efficacy in using AI for problem-solving on human autonomy: $\chi^2(1) = 5.412, p < 0.05$. For male students, self-efficacy in using AI did not significantly predict human autonomy ($\beta = 0.221, p > 0.05$). In contrast, for female students, the relationship was also non-significant but directionally negative ($\beta = -0.427, p > 0.05$).

Consistent with the full sample model, significant paths were identified from impact of using AI for problem-solving to fairness ($\beta = 0.569, p < 0.01$ for males vs. $\beta = 0.359, p < 0.05$ for females), and from meaningfulness in using AI for problem-solving to beneficence ($\beta = 0.440, p < 0.05$ for males vs. $\beta = 1.159, p < 0.01$ for females). Besides, the path from self-efficacy in using AI for problem-solving to fairness remained insignificant in the male and female groups. Self-efficacy in using AI for problem-solving was only a negative predictor for beneficence in the female group ($\beta = -0.670, p < 0.05$) but not in the male group ($\beta = -0.173, p > 0.05$).

5. Discussion

Overall, our study made the first endeavour to untangle the relationships between empowerment in using AI for problem-solving and AI ethics. We also examined gender differences in the relationship between empowerment in using AI for problem-solving and AI ethical awareness. Here are the discussions for the research questions and testing hypotheses.

Table 6

Measurement invariance of constructs of empowerment in using AI for problem-solving and AI ethics.

Models	χ^2	df	χ^2/df	CFI	TLI	RMSEA (90% CI)	SRMR	ΔCFI	$\Delta RMSEA$
Configural invariance	761.702***	310	2.457	0.965	0.957	0.069(0.063, 0.075)	0.035	—	—
Metric invariance	768.936***	324	2.373	0.965	0.959	0.067(0.061, 0.073)	0.038	0.000	-0.002
Scalar invariance	790.328 ***	338	2.338	0.965	0.960	0.066(0.060, 0.072)	0.039	0.000	-0.001

Note. CFI = Comparative fit index; TLI = Tucker-Lewis index; RMSEA = Root mean squared error of approximation; SRMR = Standardised root mean squared residual.

*** $p < 0.001$.

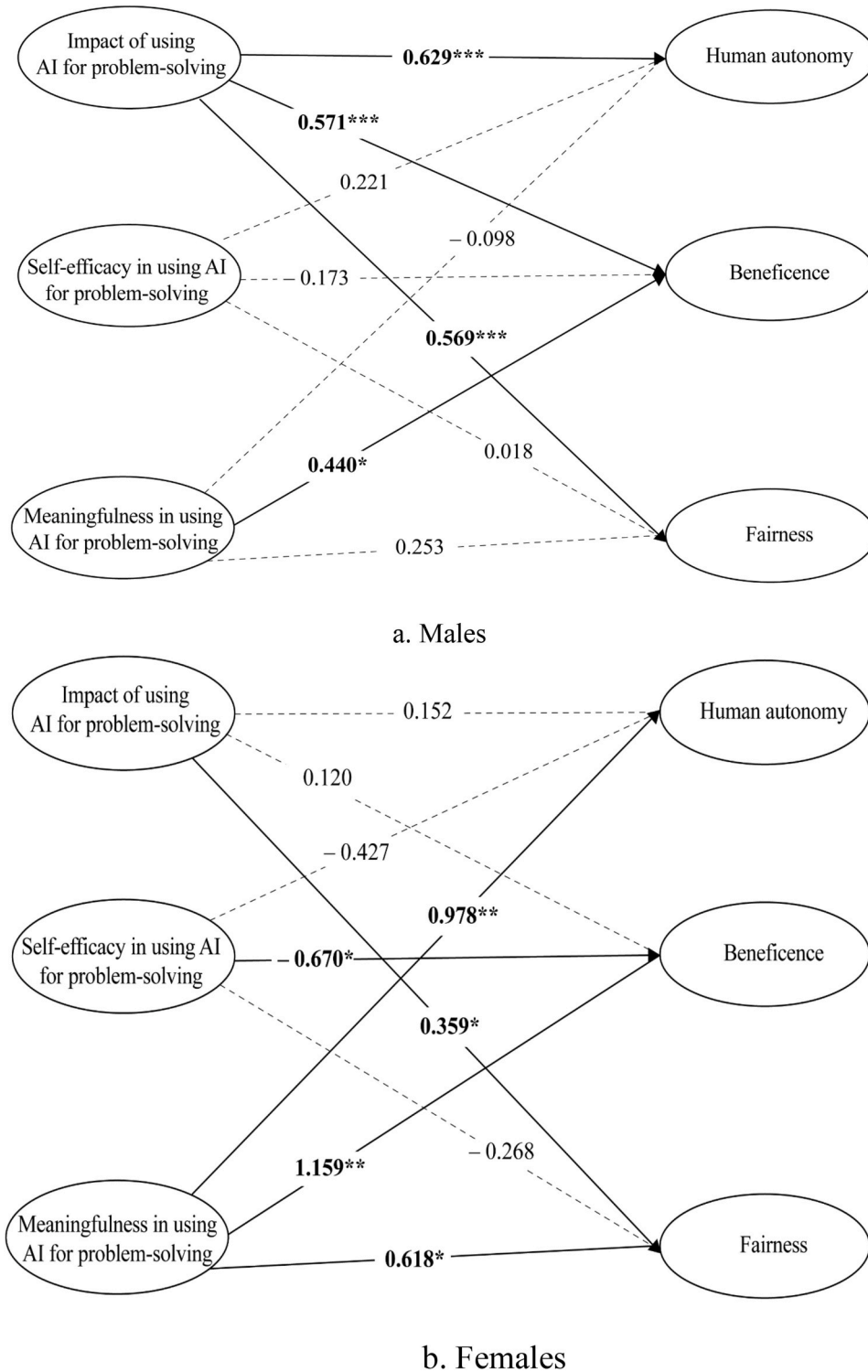


Fig. 3. Gender differences in standardised path coefficients. *Note.* Values are standardised estimates. Bold lines indicate a significant direct path; dashed lines indicate an insignificant direct path. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

5.1. Impact of using AI for problem-solving and AI ethical awareness

Our findings indicated that the impact of using AI for problem-solving significantly and positively predicted human autonomy, beneficence, and fairness. This suggests that students who feel the impact of using AI for problem-solving tend to exhibit higher intentions to consider human autonomy, beneficence, and fairness when developing their AI-based solutions. As the theories of TPB and SEVT show, attitudes

and subjective values significantly influence individuals' intentions and engagement in the chosen activities (Ajzen, 1991; Wigfield & Eccles, 2000). Another contributing factor involves the influence of significant others (e.g., peers, seniors, and teachers), as students' ethical awareness may be shaped by their observations in PBL classrooms of how peers successfully mitigate ethical risks when leveraging AI for social good (Hamid et al., 2023; Nasaj et al., 2025; Venkatesh et al., 2003).

Human autonomy has been regarded as an integral component of

Table 7

Gender differences in standardised path coefficients.

Hypotheses	χ^2	df	p-value
H1: Impact of using AI for problem-solving → human autonomy	5.062	1	0.024
H2: Impact of using AI for problem-solving → beneficence	3.041	1	0.081
H3: Impact of using AI for problem-solving → fairness	0.901	1	0.343
H4: Self-efficacy in using AI for problem-solving → human autonomy	5.412	1	0.020
H5: Self-efficacy in using AI for problem-solving → beneficence	3.273	1	0.070
H6: Self-efficacy of using AI for problem-solving → fairness	1.549	1	0.213
H7: Meaningfulness of using AI for problem-solving → human autonomy	6.243	1	0.012
H8: Meaningfulness of using AI for problem-solving → beneficence	3.226	1	0.072
H9: Meaningfulness of using AI for problem-solving → fairness	1.280	1	0.258

human-centred design and use of AI, as it is a core ethical principle that safeguards individuals from the adverse effects of technology while also grounding individuals' right to self-determination and corresponding responsibilities of other agents (Laitinen & Sahlgren, 2021). Students can acknowledge the dual impacts brought by AI through the AI literacy course that involves knowledge of AI algorithms, case studies of AI applications, real-life practices, which motivates students to strike a balance between human-led decision-making and AI-led decision-making in pursuit of human-chosen objectives (Ashok et al., 2022; Floridi & Cowls, 2019).

Furthermore, students who perceive a high impact of using AI for problem-solving tend to recognise its cognitive, affective, and social support and feel empowered to ensure their AI-based solutions yield strategic, operational, or administrative outcomes within their organisational contexts. This increases the likelihood that students will critically assess how to balance AI's benefits (e.g., providing personalised insights and data-driven recommendations, eliciting positive emotional states, and supporting shared mental models) and ethical concerns associated with AI technologies (e.g., data misuse, privacy invasion, security risks, and copyright infringement) (Jobin et al., 2019; Joksimovic et al., 2023; Kaur et al., 2022). In addition, the more strongly students perceive AI-based solutions as capable of reducing existing inequalities, expanding access to AI-driven services, and weakening stereotypes and discrimination (John-Mathews et al., 2022), the more likely they are to ensure individual, group, and counterfactual fairness in designing AI systems and developing AI-based solutions through mitigating bias during data collection, algorithm design, and user interaction (Ferrara, 2023; Jang et al., 2022; Kaur et al., 2022).

5.2. Self-efficacy in using AI for problem-solving and AI ethical awareness

In contrast to our expectation, students' self-efficacy in using AI for problem-solving was merely a significant and negative predictor of beneficence, demonstrating that students with high self-efficacy in using AI for problem-solving tend to overlook the principle of beneficence. This outcome contrasts with prior studies showing that self-efficacy is positively linked to performing beneficial behaviours and reducing unethical behaviours (Elias, 2009; Stenmark et al., 2021).

As suggested by the theories of TPB and SEVT, individuals' self-efficacy is a crucial catalyst for increasing students' behavioural intentions to use AI technology (Ajzen, 1991; Kong, Yang, & Hou, 2024; Wigfield & Eccles, 2000). However, high levels of self-efficacy may also cause students to underestimate the long-term risks of their AI-based solutions for future generations, as they often exhibit overconfidence in bridging the gap between current capabilities and future outcomes (Chiu, 2025; Vancouver & Kendall, 2006). Another possible interpretation is that self-efficacy is often associated with "personal capabilities"

and individuals' self-regulatory processes (Bandura, 1989; Seo & Ilies, 2009), so high self-efficacy students might be more performance-goal-oriented, prioritising short-term achievements and striving to complete challenging tasks. Thus, students with higher self-efficacy may prioritise time and effort to accomplish "using AI for problem-solving" projects over ethical considerations like beneficence (Halliwell et al., 2022; Moores & Chang, 2009).

5.3. Meaningfulness in using AI for problem-solving and AI ethical awareness

The results also showed that students' perceived meaningfulness in using AI for problem-solving contributed to students' considerations of the principle of beneficence when developing their AI-based solutions. Students who perceive the meaningful outcomes of using AI for solving real-life problems (e.g., reaching the overarching goal, mastering challenges, a sense of unity, amplifying human skills, good relationships with peers, and sense of belonging within an organization) may be more confident and willing to connect the potential AI-based solutions to a broader social contexts and proactively uphold their accountability for leveraging AI power for social good (Bankins & Formosa, 2023; Sadeghian et al., 2024). On the other hand, students with a stronger perception of the meaningfulness in using AI for problem-solving are characterised by more interests, commitment, involvement, and efforts to do so, which drives students to design AI projects that prioritise the continued prosperity of humanity, especially they have sufficient explainability for recognizing the impact of AI actions on society due to prior AI knowledge learning (Floridi & Cowls, 2019; Frymier et al., 1996; Kong et al., 2023).

5.4. Gender differences between empowerment in using AI for problem-solving and AI ethical awareness

Significant gender differences were observed in the associations between impact/self-efficacy/meaningfulness in using AI for problem-solving and human autonomy. Specifically, students' perceived impact of using AI for problem-solving significantly and positively predicted their consideration of human autonomy in AI-based solutions in the male group. However, this positive effect was not observed in female students. It might be because the prevailing societal construction of AI technology is as a male domain, characterised by traditional masculine attributes such as logic, structured thinking, and technical expertise (Sanders, 2005; Sultan et al., 2019). However, females often remain underrepresented in AI-related disciplines with less access to AI learning opportunities and cognitive engagement with AI technology (Jiang et al., 2024; Shams et al., 2025).

Moreover, given the entrenched gendered stereotypes in AI-flourished society and the phenomenon that most AI systems are trained on datasets disproportionately reflecting male perspectives (Cheryan et al., 2017; Cirillo et al., 2020; Faulkner, 2000), male students demonstrate significantly greater confidence and agency when interacting with AI systems. They tend to more readily discern both the advantages and limitations of AI-based solutions, enhancing their capacity to exercise human autonomy in AI-led decision-making processes.

Gender significantly moderated the relationship between self-efficacy in using AI for problem-solving and human autonomy. Nevertheless, the link between them, consistent with the full sample model, was not statistically significant in both male and female groups. This outcome supports the prior findings indicating that gender differences exist in human autonomy and students' self-efficacy in computer and social science domains (Dutra-Thomé et al., 2019; Huang, 2013; Modrek et al., 2021).

In addition, meaningfulness in using AI for problem-solving positively and significantly predicted human autonomy in the female group, but not among their male counterparts. This finding suggests that marginalized females expressed a stronger desire for increased human

autonomy when interacting with AI technologies (Dutra-Thomé et al., 2019; Modrek et al., 2021). Females tend to exhibit extensive social engagement and heightened sensitivity to social stimuli and interpersonal dynamics; consequently, they tend to prioritise communal relationships, including those with family, peers, and the broader community (Antonucci & Akiyama, 1987; Cole, 2004; Thomas, 2011). Unfortunately, females have frequently been found to experience lower levels of cognitive engagement and confidence in AI-related fields (Diekmann et al., 2017; Jiang et al., 2024; Shams et al., 2025). Thus, they are inclined to perceive the communal benefits of applying AI to address societal challenges, while simultaneously recognizing their systemic underrepresentation in AI-related domains. This dual awareness appears to motivate stronger assertions of human autonomy when engaging with AI systems to foster greater inclusivity within the field.

There were no significant gender differences among the remaining relationships between empowerment in using AI for problem-solving and AI ethics. Some relationships were still significant, which is consistent with the full sample model (e.g., Impact of using AI for problem-solving → fairness; meaningfulness in using AI for problem-solving → beneficence) in the male and female groups. Others were not significant anymore in the subgroup (e.g., Impact of using AI for problem-solving → beneficence; self-efficacy in using AI for problem-solving → beneficence). It is unclear the exact nature of these findings, possibly due to the high within-gender variability or insufficient subgroup sample size, or the interaction effect between gender with unidentified factors (e.g., prior AI experience, disciplines).

6. Implications, limitations, future directions, and conclusions

This study filled a gap by integrating psychological empowerment as an essential learner characteristic within the AI ethics framework and by providing empirical evidence of the significant effect of students' impact, self-efficacy, and meaningfulness on their ethical awareness at the aspects human autonomy, beneficence, and fairness in AI environments. The findings demonstrated that the impact of using AI for problem-solving has a positive influence on human autonomy, beneficence, and fairness, and meaningfulness in using AI for problem-solving positively predicted beneficence. Nevertheless, self-efficacy in using AI for problem-solving negatively predicted beneficence. Gender also significantly moderated the relationships between human autonomy and impact of using AI for problem-solving, self-efficacy in using AI for problem-solving, and meaningfulness in using AI for problem-solving.

The findings of this study provide the following implications regarding AI applications and AI ethics education. Theoretically, this study expanded the applicability of the TPB, SEVT, and UTAUT models in the AI context. It also highlights the importance of providing supportive environments (e.g., facilitating conditions) that involve using AI for problem-solving in real-life scenarios with all elements of these models integrated, such as attitudes, subjective task values, and performance expectancy, and perceived behavioural control (Ajzen, 1991; Eccles & Wigfield, 2020; Venkatesh et al., 2003).

Secondly, the findings of this study echo prior research, showing that specific empowerment courses serve as an effective intervention to develop students' ethical awareness by engaging them in social practices of using AI for problem-solving until normativity is formed (Abbasi et al., 2019; Bagherzadeh et al., 2021; Corrêa et al., 2024). Furthermore, the results support Aristotle's argument that both practices and mentorship are crucial catalysts in facilitating students' understanding of ethical principles and decision-making (Begley, 2006).

From the pedagogical perspective, it is essential for educators and practitioners to consider how psychological empowerment in using AI for problem-solving could be integrated into AI ethics education to increase students' motivations for ethical use of AI technology in the AI-saturated world. They should prioritise the development and implementation of empowerment programmes that enable students to recognise their engagement impact on their social community, find

meaning in the problem-solving tasks, and increase their confidence in completing them (Gansser & Reich, 2021; Lysova et al., 2019; Stenmark et al., 2021). This involves designing collaborative and agency-oriented activities, providing appropriate levels of challenge and support, and fostering a positive and collaborative classroom environment. In addition, curriculum designers and administrators should consider gender bias and discrimination when developing empowerment programmes for AI applications and incorporating AI ethics education, with the aim of promoting greater inclusivity in AI learning environments.

As for instructors, they should apply the PBL approach to create facilitating conditions that enable students to experience the use of AI to solve problems in their real-life scenarios and solve problems at the cognitive, affective, and social levels (Dwivedi et al., 2019; Joksimovic et al., 2023). Through this experiential learning process, students develop self-efficacy in utilising AI tools and metacognition of AI's meaningfulness and impacts on their learning processes and daily lives. Besides, students in the PBL context could observe how important and supportive others are in making ethical considerations when using AI technology, promoting their shared beliefs for the positive impacts and meaningfulness for using AI for problem-solving, thereby strengthening students' intentions to engage more actively with generative AI tools in ways that align with ethical norms (Bairaktarova & Woodcock, 2017; Tseng et al., 2022).

From the perspective of students, they are encouraged to actively engage in collective problem formulation, uphold inquiry, and build their ethical awareness through bridging the principle-practice gap of AI ethics (Almazroui, 2023; Corrêa et al., 2024; Hamid et al., 2023). Students also need to share successful experiences with their counterparts regarding how they use AI tools to tackle various problems while making ethical considerations throughout the process (Blumenfeld et al., 1991). Such sharing can enhance their recognition of the meaningfulness of AI applications and the significance of ethical awareness when using AI for solving real-life problems.

Some limitations are warranted for consideration. Firstly, this study is dependent on self-reported data with a cross-sectional design. Future studies could use longitudinal designs to track changes over time or supplement qualitative methods (e.g., experimental design, observation, interview, or stimulated recall) to offer a richer and profound understanding of the dynamics between students' empowerment in using AI for problem-solving and AI ethical awareness (Creswell & Clark, 2017).

Secondly, caution should be taken regarding the generalizability of the results in other contexts since our diverse sample was based on the Hong Kong students. Students in Hong Kong operate within a Confucian Heritage Culture (CHC) context featured by interdependence, higher power distance, and an analytic mode of thought. In contrast, their Anglo-Saxon counterparts typically exhibit individualist values characterised by independence, individual initiative, low power distance, and intuitive approaches. These contextual disparities influence students' perception of empowerment in using AI for problem-solving (e.g., what competence is necessary to use AI for problem-solving) and AI ethics (e.g., data privacy and AI in immigration control) across regions, such as North America, Europe, and East Asia (Kong et al., 2025; 2025h; Eigeartaigh et al., 2020). Hence, future research may involve students from different regions to examine whether contextual differences impact the link between students' empowerment in using AI for problem-solving and AI ethics.

Lastly, there may exist mediation or moderation effects that lead to the insignificant links or gender differences in some paths. Hence, future research can explore how potential mediators (e.g., student engagement) or gender-related moderators (e.g., AI educational experience) (Jang et al., 2022; Nguyen et al., 2024) influence the link between empowerment in using AI for problem-solving and AI ethical awareness.

CRedit authorship contribution statement

Siu Cheung Kong: Conceptualization, Data curation, Funding

acquisition, Project administration, Resources, Supervision, Writing – review & editing. **Jinyu Zhu:** Formal analysis, Investigation, Methodology, Writing – original draft.

Ethical approval

The submitted work is original and has not been published elsewhere in any form or language, partially or in full. The author has included the ethical approval and all the relevant declaration statements in the submission of this manuscript.

Informed consent: The author has obtained approval by the ethics committee of the University for the research involving humans and informed consent of human participants of this study.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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